

Group 5: Air Quality Patterns in African Cities

Phased Implementation Plan

Client: United Nations Environment Programme (UNEP)

API: OpenAQ v3

Team: Samuel Tokoye, Paul Kibet Miningwa, Winnie Odoyo, Gloria Simiyu Wandabwa

Cities: Nairobi, Kampala, Kigali, Addis Ababa, Johannesburg, Lagos

Concurrency guide: what must finish first

The pipeline has one real bottleneck. Everything before it must happen in strict order, one person's output feeding the next. Everything after it can split across the team.

Must be done sequentially, in this order, before real parallel work starts:

1. **Phase 0 (Setup):** repo, API keys, folder structure. Nobody can pull data without this.
 2. **Phase 1 (Acquisition):** raw JSON per city must exist before anyone can assess it.
 3. **Phase 2 (QA):** must run on real raw data, and its findings (which cities/stations are usable, what needs dropping) directly shape cleaning decisions in Phase 3.
 4. **Phase 3 (Cleaning):** produces the master CSV. Stats, charts, and the dashboard all read from this one file, building any of them on a guessed schema risks rework once the real cleaned data lands.
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Phase 0: Setup & Foundations

Goal: Everyone can run the project locally and access shared infrastructure.

Task	Owner
Register individual OpenAQ API keys at explore.openaq.org	All
Create shared GitHub repo (<code>group5-air-quality</code>), add all members as collaborators	Gloria
Set up repo structure: <code>/notebooks</code> , <code>/data/raw</code> , <code>/data/cleaned</code> , <code>/dashboard</code> , <code>/slides</code>	Gloria
Confirm final 6 cities and pollutant scope (PM2.5 primary, others as available)	All
Quick group call: assign Phase-owner roles below	All

Deliverable: Working repo, everyone able to `git clone` and run a test API call.

Phase 1: Data Acquisition

Owner: Samuel (lead) | **Goal:** Raw JSON pulled and saved for all 6 cities.

- Use OpenAQ v3 `/locations` endpoint (bounding box per city) to find monitoring stations
- Use `/sensors` and `/measurements` (or averaged hourly/daily endpoints) to pull PM2.5 and other available pollutants for the most recent 3–5 years
- Document every endpoint, parameter, and filter used (required for the brief's documentation requirement)
- Save raw responses as JSON per city: `data/raw/{city}_raw.json`
- Flag early: any city with sparse/no station coverage (common for some African cities on OpenAQ), decide fallback (drop city vs. use best available proxy) before Phase 2 starts

Deliverable: 6 raw JSON files + an API documentation section (endpoints/params) for the notebook.

Phase 2: Data Quality Assessment

Owner: Gloria | **Goal:** Documented, evidence-based picture of data health before any cleaning happens.

- Missing values: % missing per city/pollutant/time period

- Duplicate records: check for repeated timestamps per station
- Outliers: flag implausible PM2.5 readings (e.g., sensor error spikes)
- Completeness: which stations have near-continuous coverage vs. gaps
- Consistency: units, timezones, averaging periods aligned across cities

Deliverable: QA section of notebook with tables/summary stats — this stays *visible* in the final notebook, not folded silently into cleaning.

Phase 3: Cleaning & Preprocessing

Owner: Winnie | **Goal:** Analysis-ready cleaned CSV.

- Handle missing values (interpolation vs. exclusion: document the reasoning)
- De-duplicate, standardize timestamps/timezones and units
- Merge per-city files into one master DataFrame with a `city` column
- Write a short "preprocessing decisions" log: this doubles as content for the notebook

Deliverable: `data/cleaned/air_quality_master.csv`

Phase 4: EDA & Statistical Analysis

Owner: Paul | **Goal:** Answer the 6 client questions with ≥ 3 statistical techniques.

Map techniques to questions:

- **Descriptive/ranking** → Q1 (highest average PM2.5), Q2 (most-monitored pollutants)
- **Trend/time-series analysis** → Q3 (seasonal variation)
- **Threshold/exceedance-rate analysis** → Q4 (% of readings above WHO guideline)
- **Data-completeness scoring** → Q5 (most reliable monitoring stations)
- Synthesize all of the above → Q6 (prioritization recommendation)

Deliverable: EDA + stats section of notebook, one clear answer per client question with supporting numbers.

Phase 5: Visualizations (≥5 charts)

Owner: Gloria | **Goal:** Each chart directly answers a question, with a justification line.

Suggested set:

1. Bar chart: average PM2.5 by city (Q1)
2. Heatmap/grouped bar: pollutant monitoring frequency by city (Q2)
3. Line/box plot: seasonal PM2.5 variation per city (Q3)
4. Stacked bar or line: % readings exceeding WHO guideline over time (Q4)
5. Bar/scatter: data completeness score by station (Q5)
6. (Optional 6th) Map: station locations colored by pollution level

Deliverable: Finalized chart set + one-sentence justification per chart, ready to drop into notebook and dashboard.

Phase 6: Interactive Dashboard (PowerBI)

Owner: Winnie | **Goal:** Meets all brief requirements

- 3 KPI cards: overall avg PM2.5, most-polluted city, % exceeding WHO guideline
- 4 interactive charts (from Phase 5, made reactive to filters)
- Filters: city (multiselect), pollutant, date range
- Map: station points colored by PM2.5 level
- Executive summary block (markdown, written once Phase 7 findings are ready)
- Drill-down: filters narrow both KPIs and charts together

Deliverable: PowerBI dashboard.

Phase 7: Findings & Recommendations

Owner: Samuel (with input from all) | **Goal:** Evidence-backed answers for UNEP.

- Draft direct answers to all 6 client questions, each citing the specific stat/chart that supports it

- Draft prioritization recommendation (which cities need intervention, and why)
- Write the executive summary (used in both dashboard and slides)

Deliverable: Findings & recommendations write-up, feeds directly into slides and dashboard summary.

Phase 8: Packaging & Slides

Owner: Winnie (slides) + Samuel (repo/README) | **Goal:** All 6 deliverables submission-ready.

- Finalize Jupyter notebook (narrative + code, run top-to-bottom cleanly)
- Confirm all 6 deliverables present: notebook, raw JSON, cleaned CSV, dashboard, slides, GitHub repo
- Build 10-minute presentation deck (problem → method → findings → recommendations)
- Write README with setup instructions and each member's documented contribution

Deliverable: Complete submission package.
